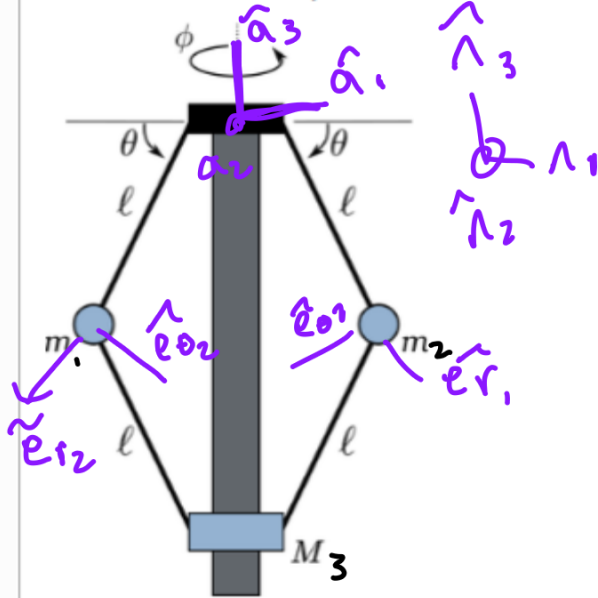


Problem 1

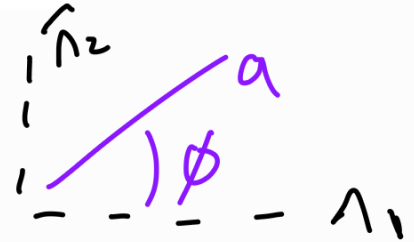
A flyball governor is constructed of 2 balls of mass m located at the end of a massless bar of length ℓ , each of which is attached to a freely-sliding collar of mass M by another massless bar of length ℓ . Define the angle of the top bar down from the horizontal as θ and the rotation of the entire assembly around the vertical axis as the angle ϕ .



$$\hat{a}_1 = \cos\theta \hat{\lambda}_1 + \sin\theta \hat{\lambda}_2$$

$$\hat{a}_3 = \hat{\lambda}_3$$

$$\hat{a}_2 = \hat{a}_3 \times \hat{a}_1$$



a) $\vec{r}_1 = L \hat{e}_r$

$$\omega_{A/N} = \dot{\phi} \hat{\lambda}_3$$

$$\hat{e}_{r'} = \cos\theta \hat{a}_1 - \sin\theta \hat{a}_3$$

$$\hat{e}_{\theta'} = -\sin\theta \hat{a}_1 - \cos\theta \hat{a}_3$$

$$\omega_{E1/N} = \dot{\phi} \hat{a}_3 + \dot{\theta} \hat{a}_2$$

$$\omega_{E2/N} = \dot{\phi} \hat{a}_3 - \dot{\theta} \hat{a}_2$$

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$$\vec{r}_1 = L \hat{e}_{r^1}$$

$$\vec{r}_2 = L \hat{e}_{r^2}$$

$$\vec{r}_M = -2L \sin\theta \hat{n}_3$$

$$\frac{d\vec{r}_1}{dt} = \frac{d\vec{r}_1}{dt} + \vec{\omega}_{E/N} \times \vec{r}_1$$

$$\dot{\theta} \hat{a}_3 + \dot{\theta} \hat{a}_2 \times L \hat{e}_{r^1}$$

$$= (\dot{\theta} \hat{a}_3 + \dot{\theta} \hat{a}_2) \times (L \cos\theta \hat{a}_1 - L \sin\theta \hat{a}_3)$$

$$L \cos\theta \dot{\theta} \hat{a}_2 + \vec{0} - L \cos\theta \dot{\theta} \hat{a}_3$$

$$- L \sin\theta \dot{\theta} \hat{a}_1$$

$$\therefore \vec{V}_1 = L \cos\theta \dot{\theta} \hat{a}_2 + L \dot{\theta} \hat{e}_{\theta^1}$$

due to symmetry,

$$\vec{V}_2 = -L \cos\theta \dot{\theta} \hat{a}_2 + L \dot{\theta} \hat{e}_{\theta^2}$$

$$\vec{r}_{m3} = -zL \sin\theta \hat{n}_3$$

$$\frac{d\vec{r}_{m3}}{dt} = -zL \cos\theta \dot{\theta} \hat{n}_3$$

b) θ, ϕ = minimum set

$$\theta_1, \phi_1, \theta_2, \phi_2, z_{m3} =$$

redundant set

c) $\vec{V}_1 = L \dot{\theta} \hat{e}_\theta + L \cos\theta \dot{\phi} \hat{a}_2$

$$\vec{V}_2 = L \dot{\theta} \hat{e}_\theta - L \cos\theta \dot{\phi} \hat{a}_2$$

$$\vec{V}_m = -zL \cos\theta \dot{\theta} \hat{n}_3$$

$$\gamma_{1\theta} = L \hat{e}_\theta$$

$$\gamma_{2\theta} = L \hat{e}_\theta^2$$

$$\gamma_{\mu\theta} = -2L \cos\theta \hat{\Lambda}_3$$

$$\gamma_{1\phi} = L \cos\theta \hat{a}_z$$

$$\gamma_{2\phi} = -L \cos\theta \hat{a}_z$$

$$\gamma_{\mu\phi} = \vec{0}$$

d)

$$|\vec{V}_1|^2 = L^2 \dot{\theta}^2 + L^2 \dot{\phi}^2 \cos^2\theta$$

$$|\vec{V}_2|^2 = L^2 \dot{\theta}^2 + L^2 \dot{\phi}^2 \cos^2\theta$$

$$|\vec{V}_M|^2 = 4L^2 \dot{\theta}^2 \cos^2\theta$$

$$T = \frac{1}{2} M \cdot 4L^2 \dot{\theta}^2 \cos^2\theta$$

$$+ m \cdot (L^2 \dot{\theta}^2 + L^2 \dot{\phi}^2 \cos^2\theta)$$

$$T = 2ML^2\dot{\theta}^2\cos^2\theta + mL^2\dot{\theta}^2 + mL^2\dot{\phi}^2\cos^2\theta$$

$$V = -2mgL\sin\theta$$

$$mg(2L\sin\theta)$$

$$\therefore V = -2gL\sin\theta(M+m)$$

$$L = \cancel{m}L^2\dot{\theta}^2 + mL^2\dot{\phi}^2\cos^2\theta$$

$$+ 2\cancel{M}L^2\dot{\theta}^2\cos^2\theta$$

$$+ 2gL\sin\theta(M+m)$$

$$L = L^2(m + 2M\cos^2\theta)\dot{\theta}^2$$

$$+ mL^2\cos^2\theta\dot{\phi}^2$$

$$+ 2(m+M)gL\sin\theta$$

$$\theta: \frac{\partial L}{\partial \dot{\theta}} = 2L^2 (m + 2M \cos^2 \theta) \dot{\theta}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = 2L^2 (m + 2M \cos^2 \theta) \ddot{\theta} - 8ML^2 \dot{\theta}^2 \sin \theta \cos \theta$$

$$\begin{aligned} \frac{\partial L}{\partial \theta} = & -4ML^2 \dot{\theta}^2 \sin \theta \cos \theta \\ & 2mL^2 \dot{\theta}^2 \sin \theta \cos \theta \\ & + \\ & 2(m + M)gL \cos \theta \end{aligned}$$

divide
by $2L^2$

$$\begin{aligned} & (m + 2M \cos^2 \theta) \ddot{\theta} - 2M \dot{\theta}^2 \sin \theta \cos \theta \\ & + \\ & m \dot{\theta}^2 \sin \theta \cos \theta \\ & - \\ & (m + M) \frac{g}{L} \cos \theta = 0 \end{aligned}$$

$$\emptyset: \frac{\partial L}{\partial \emptyset} = 0$$

$$\frac{\partial L}{\partial \dot{\emptyset}} = 2ML^2 \cos^2 \theta \dot{\emptyset}$$

$$\frac{d}{dt} (2ML^2 \cos^2 \theta \dot{\emptyset}) = 0$$

$$2ML^2 \cos^2 \theta \ddot{\emptyset} - 4ML^2 \dot{\theta} \dot{\emptyset} \sin \theta \cos \theta$$

$$\ddot{\emptyset} = \frac{4ML^2 \sin \theta \cos \theta \dot{\theta} \dot{\emptyset}}{2ML^2 \cos^2 \theta}$$

$$\ddot{\emptyset} = 2 \dot{\theta} \dot{\emptyset} \tan \theta$$

B/c time constant, T is equal

c)

$$H = T + V$$

$$P_\theta = \frac{\partial L}{\partial \dot{\theta}} = 2L^2 (m + 2M \cos^2 \theta) \dot{\theta}$$

$$P_\phi = \frac{\partial L}{\partial \dot{\phi}} = 2ML^2 \cos^2 \theta \dot{\phi}$$

$$\dot{\theta} = \frac{P_\theta}{2L^2 (m + 2M \cos^2 \theta)}$$

$$\dot{\phi} = \frac{P_\phi}{2ML^2 \cos^2 \theta}$$

$$H = 2L^2 (m + M \cos^2 \theta) \dot{\theta}^2 +$$

$$ML^2 \cos^2 \theta \dot{\phi}^2$$

$$- 2gL \sin \theta (m + M)$$

$\therefore$

$$H = \frac{p_\theta^2}{4L^2(m + 2M\cos^2\theta)} + \frac{p_\phi^2}{4ML^2\cos^2\theta} - 2(M+m)gL\sin\theta$$

$$\dot{\theta} = \frac{\partial H}{\partial p_\theta} = \frac{p_\theta}{2L^2(m + 2M\cos^2\theta)}$$

$$\dot{\phi} = \frac{\partial H}{\partial p_\phi} = \frac{p_\phi}{2ML^2\cos^2\theta}$$

$$\begin{aligned} \dot{p}_\theta = -\frac{\partial H}{\partial \theta} = & -\frac{Mp_\theta^2 \sin\theta \cos\theta}{L^2(m + 2M\cos^2\theta)^2} \\ & - \frac{p_\phi^2 \sin\theta}{2ML^2\cos^3\theta} + 2(M+m)g\cos\theta \end{aligned}$$

$$\dot{p}_\phi = -\frac{\partial H}{\partial \phi} = 0$$

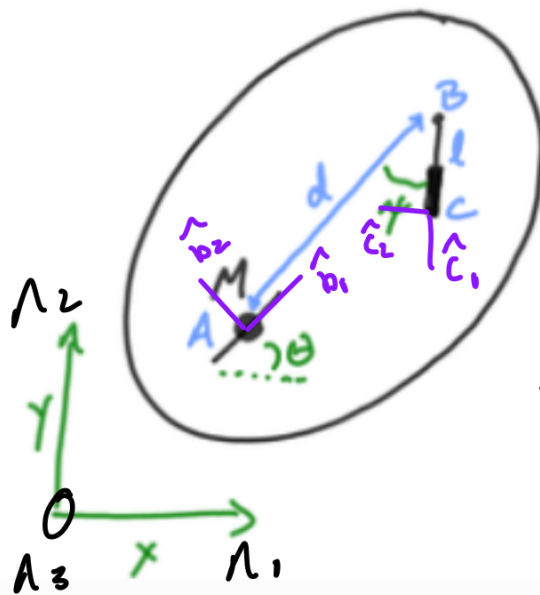
The rest of the questions
were answered in latex

Problem 2

For your second problem, consider the motion of the Razer Crazy Cart Shuffle. Kids make the Crazy Cart go by moving the bar on the front back and forth with their feet, meaning that it works entirely through nonholonomic constraints.



The center of mass is centered on the knife-edge constraint at A , and its location in an inertial frame is given by x, y . θ represents the heading of the cart and ψ represents the angle between the front wheel to the heading of the cart. A second knife-edge constraint is located on the wheel (point C) at a distance ℓ behind the pivot point B . The pivot point B is located directly in line with the point A a distance d ahead. Treat the front wheel as massless.



$$\begin{aligned} \hat{b}_2 &= -\sin\theta \hat{n}_1 + \cos\theta \hat{n}_2 \\ \hat{c}_2 &= -\sin\psi \hat{b}_1 - \cos\psi \hat{b}_2 \\ &= (-\sin\psi \cos\theta - \sin\psi \sin\theta) \hat{n}_1 \\ &\quad + (-\cos\psi \sin\theta - \cos\psi \cos\theta) \hat{n}_2 \end{aligned}$$

1)

$${}^N V_A = \dot{x} \hat{x} + \dot{y} \hat{y}$$

$$\text{constraint: } {}^N V_A \cdot \hat{b}_2 = 0$$

$$\phi_1 = -\dot{x} \sin\theta + \dot{y} \cos\theta = 0$$

C:

$$\vec{r}_C = \vec{r}_A + d \hat{b}_1 - L \hat{C}_1$$

$${}^N \omega^B = \dot{\theta} \hat{\Lambda}_3$$

$${}^N \omega^C = (\dot{\theta} + \dot{\psi}) \hat{\Lambda}_3$$

$$\frac{{}^N d \hat{b}_1}{dt} = \dot{\theta} \hat{\Lambda}_3 \times \hat{b}_1 = \dot{\theta} \hat{b}_2$$

$$\frac{{}^N d \hat{C}_1}{dt} = (\dot{\theta} + \dot{\psi}) \hat{\Lambda}_3 \times \hat{C}_1 = (\dot{\theta} + \dot{\psi}) \hat{C}_2$$

$${}^N \vec{V}_C = \frac{d \vec{r}_A}{dt} + \frac{{}^N d (d \hat{b}_1)}{dt} - \frac{{}^N d (L \hat{C}_1)}{dt}$$

$$= \dot{x} \hat{\Lambda}_1 + \dot{y} \hat{\Lambda}_2 + d \dot{\theta} \hat{b}_2 - L (\dot{\theta} + \dot{\psi}) \hat{C}_2$$

b/c knife edge $\vec{V}_C \cdot \hat{C}_2 = 0$

$$N \vec{V}_A \cdot \hat{C}_2 + d \dot{\theta} \cos \psi - L(\dot{\theta} + \dot{\psi}) = 0$$

$$N \vec{V}_A \cdot \hat{C}_2 :$$

$$- \dot{x} \sin(\theta + \psi) + \dot{y} \cos(\theta + \psi)$$

$$= \quad \quad \quad \cancel{\dot{x}} = 0$$

$$\cos \psi (-\dot{x} \sin \theta + \dot{y} \cos \theta)$$

$$- \sin \psi (\dot{x} \cos \theta + \dot{y} \sin \theta)$$

$$\begin{aligned} \therefore \\ \phi_2 = & -\sin \psi (\dot{x} \cos \theta + \dot{y} \sin \theta) \\ & + \\ & (d \cos \psi - L) \dot{\theta} - L \dot{\psi} \end{aligned}$$

b)

$$V = 0$$

$$T = \frac{1}{2} M (\dot{x}^2 + \dot{y}^2) + \frac{1}{2} I \dot{\theta}^2$$

$$L = T - V$$

$$L = \frac{1}{2} M (\dot{x}^2 + \dot{y}^2) + \frac{1}{2} I \dot{\theta}^2$$

$$\mathcal{O}_j = \sum_i a_{ji} \ddot{q}_i + b_j = 0$$

coords are x, y, θ (ψ is rider input)

$$a_{1x} = -\sin\theta$$

$$a_{2x} = -\sin\psi \cos\theta$$

$$a_{1y} = \cos\theta$$

$$a_{2y} = -\sin\psi \sin\theta$$

$$a_{1\theta} = 0$$

$$a_{2\theta} = d \cos\psi - L$$

$$x: M \ddot{x} = -\lambda_1 \sin \theta - \lambda_2 \sin \psi \cos \theta$$

$$y: M \ddot{y} = \lambda_1 \cos \theta - \lambda_2 \sin \psi \sin \theta$$

$$\theta: I \ddot{\theta} = \lambda_2 (\sin \psi \cos \theta - 1)$$

$$c) \text{ let } v = \dot{x} \hat{b}_1$$

$$v = \dot{x} \cos \theta + \dot{y} \sin \theta \quad \text{solve}$$

$$0 = -\dot{x} \sin \theta + \dot{y} \cos \theta$$

$$\therefore \dot{x} = v \cos \theta$$

$$\dot{y} = v \sin \theta$$

$$\therefore \ddot{x} = \dot{v} \cos \theta - v \dot{\theta} \sin \theta$$

$$\ddot{y} = \dot{v} \sin \theta + v \dot{\theta} \cos \theta$$

=

$$\vec{a}_A = \dot{v} \hat{b}_1 + v \dot{\theta} \hat{b}_2$$

$$m^N \vec{a}_A = \lambda_1 (-\sin\theta \hat{e}_1 + \cos\theta \hat{e}_2)$$

$$\lambda_2 \sin\psi (\cos\theta \hat{e}_1 + \sin\theta \hat{e}_2)$$

$$\therefore m^N \vec{a}_A = \lambda_1 \hat{e}_2 - \lambda_2 \sin\psi \hat{e}_1$$

$$m^N \vec{a}_A \cdot \hat{e}_1 = -\lambda_2 \sin\psi$$

$$= m \dot{V} = -\lambda_2 \sin\psi$$

$$m^N \vec{a}_A \cdot \hat{e}_2 = \lambda_1$$

$$\therefore \lambda_1 = m v \dot{\theta}$$

...

$$m \dot{v} = -\lambda_2 \sin \psi$$

$$m v \dot{\theta} = \lambda_1$$

$$I \ddot{\theta} = \lambda_2 (d \cos \psi - L)$$

$$\phi_{2,1} \quad -v \sin \psi + (d \cos \psi - L) \dot{\theta} - L \dot{\psi} = 0$$

$$\hookrightarrow \ddot{\theta} = \frac{L \dot{\psi} + v \sin \psi}{d \cos \psi - L}$$

$\therefore$

$$\lambda_1 = m v \cdot \frac{L \dot{\psi} + v \sin \psi}{d \cos \psi - L}$$

$$\ddot{\theta} = \frac{(\dot{v} \sin \psi + v \dot{\psi} \cos \psi + L \ddot{\psi})(d \cos \psi - L)}{(v \sin \psi + L \dot{\psi})(-d \sin \psi \dot{\psi})}$$

$$\lambda_2 = \frac{I \ddot{\theta}}{d \cos \psi - L}$$

$$\therefore \dot{V} = -\frac{I \sin \psi}{(m(d \cos \psi - L))} \ddot{\theta} \quad \swarrow$$

$$\ddot{\theta} = \frac{(d \cos \psi - L)(V \dot{\psi} \cos \psi + L \ddot{\psi}) + d \dot{\psi} \sin \psi (V \sin \psi + L \ddot{\psi})}{(d \cos \psi - L)^2 + \frac{I}{m} \sin^2 \psi}$$

$$\lambda_2 = \frac{I}{d \cos \psi - L}$$

$$\frac{(d \cos \psi - L)(V \dot{\psi} \cos \psi + L \ddot{\psi}) + d \dot{\psi} \sin \psi (V \sin \psi + L \ddot{\psi})}{(d \cos \psi - L)^2 + \frac{I}{m} \sin^2 \psi}$$

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